The Noisy Work of Uncertainty Visualisation Research: A Review

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Abstract

Uncertainty visualisation is quickly becoming a hot topic in information visualisation. Existing reviews in the field take the definition and purpose of an uncertainty visualisation to be self evident which results in a large amout of conflicting information. This conflict largely stems from a conflation between uncertainty visualisations designed for decision making and those designed to prevent false conclusions. We coin the term "signal suppression" to describe a visualisation that is designed for preventing false conclusions, as the approach demands that the signal (i.e. the collective take away of the estimates) is suppressed by the noise (i.e. the variance on those estimates). We argue that the current standards in visualisation suggest that uncertainty visualisations designed for decision making should not be considered uncertainty visualisations at all. Therefore, future work should focus on signal suppression. Effective signal suppression requires us to communicate the signal and the noise as a single "validity of signal" variable, and doing so proves to be difficult with current methods. We illustrate current approaches to uncertainty visualisation by showing how they would change the visual apprearance of a choropleth map. These maps allow us to see

why some methods succeed at signal suppression, while others fall short. Evaluating visualisations on how well they perform signal suppression also proves to be difficult, as it involves measuring the effect of noise, a variable we typically try to ignore. We suggest authors use qualitative studies or compare uncertainty visualisations to the relevant hypothesis tests.

1 Introduction

The phrase "uncertainty visualisation" can sometimes feel like a bit of an oxymoron. Uncertainty is often discussed as though it is an omnipresent spectre, touching every stage of our analysis despite the fact that we don't seem to have a precise definition of what it is. A point that is either directly stated by authors who work in the field (Spiegelhalter 2017, Griethe & Schumann 2006) or implied by describing a myriad of things uncertainty could be without a concrete definition (Kinkeldey et al. 2014, Hullman 2016). Contrary to the murky waters of uncertainty, visualisations illuminates and unveils the unknowns in our data. This is illustrated with datasets such as Anscombe's quartet (Anscombe 1973) or the Datasaurus Dozen (Matejka & Fitzmaurice 2017, Locke & D'Agostino McGowan 2018) where visual statistics reveal elements of the data that are hidden to the typical summary statistics. This leads us to the conclusions that uncertainty visualisation is about revealing the unknown and shining a light on the more questionable parts of our analysis.

This sentiment is regularly echoed by the literature where we see authors express that visualising uncertainty is necessary for transparency and improved decision making (Padilla et al. 2022). Some authors even go so far as to claim that failing to include uncertainty is akin to fraud or lying (Hullman 2020, Manski 2020). There is good reason to believe these assertions are true and failing to incorporate the uncertain information in an analysis will lead to research being undermined, uninformative, or untrustworthy. A 6-month survey of anti-mask groups on Facebook during the COVID-19 pandemic showed that anti-maskers made persuasive arguments by exploiting the uncertainty information that was ignored by pro-maskers (**?**). Failing to present uncertain information can lead to poor estimation of failure probabilities (**?**), as well as poorer decision making both experimentally (**??**Kay et al. 2016, Fernandes et al. 2018) and in practice (**?**). We see authors repeatedly suggest

uncertainty is important to include as it communicates the legitimacy (or illegitimacy) of the conclusion drawn from visual inference (Correll & Gleicher 2014, Kale et al. 2018, Griethe & Schumann 2006). Despite this, a survey conducted by Hullman (2020) found that very few authors actually include uncertainty in their visualisations, with only a quarter of respondents included uncertainty in 50% or more of their visualisations. Part of the abstinence of uncertainty visualisation comes from the fact that authors think that the plots will be overwhelming or they are uncertain how to calculate the uncertainty (Hullman 2020).

When reading the uncertainty visualisation literature itself, we can see why people will think the plots are overwhelming, as the field of uncertainty visualisation itself is quite overwhelming. The literature often reads like several independent conversations happening at once, largely because for many years, that is what the literature was. Early mentions of uncertainty visualisation start to spring up in the late 1980s (Ibrekk & Morgan 1987) as a discussion in the most effective ways to communicate random variables and their associated density. This work was motivated by conversations around risk communication, a subfield of psychology where heuristics, such as risk or ambiguity aversion, make it difficult to communicate probabilities to the public. This risk communication sub field end up with their own version of "uncertainty visualisation" that is not as concerned with conveying the correct shape and sense of a distribution, but rather in designing visualisations that help the public make accurate informed decisions about things like the outcomes of medical procedures (Spiegelhalter 2017). In parallel, around the early 1990s, authors in geoscience declare uncertainty to be an essential aspect to any information display (MacEachren 1992, Carr et al. 1992) and start working towards implementing it in their visualisations. Since geoscience often involves high dimensional (usually at least 4 axis) data with inherent

measurement error, getting all the information into a single display was proving difficult, even to this day (Peña-Araya et al. 2025). Thrown into the melting pot and stewed for years, these separate motivations behind uncertainty visualisation all bleed together and are treated interchangeable as early work in"uncertainty visualisation".

This "melting pot of ideas" becomes a bit of a problem when we try and write review papers that turn the literature into a set of generalised theoretical principles. Rather than providing an overarching, cohesive narrative of the field, it is far more common to see a review paper lament the *difficulties* faced when trying to extract any cohesive insights. Kinkeldey et al. (2014) found most experimental methods to be ad hoc, with no commonly agreed upon methodology, formalisations, or greater goal of describing general principals. Hullman (2016) noticed there is a serious noise issue in the field, with noise from participants misunderstanding visualisations, misinterpreting questions, and incorrectly applying heuristics all overwhelming any information we can glean from studies. MacEachren et al. (2005) identified so much contradicting evidence that they were able to spend an entire page discussing the contradictions around a question as simple as "Should I map uncertainty to colour hue?". Griethe & Schumann (2006) did not identify common themes, but instead listed the findings and opinions of a collection of papers. Padilla et al. (2022) was able to synthesise some results cognative effects that arise in uncertainty visualisation experiments, however these effects were each discussed in isolation as a list of considerations an author might make rather than an overarching theory of rules for effective uncertainty visualisation. While these reviews are thorough in scope, none discuss how the existing literature contribute to the broader goal of uncertainty visualisation. The problem faced by the literature is easily summarised with a famous quote by Henri Poincaré.

> "Science is built up of facts, as a house is built of stones; but an accumulation

of facts is no more a science than a heap of stones is a house." - Henri Poincaré (1905)

That is to say, despite the wealth of reviews, the field of uncertainty visualisation remains a heap of stones. This review attempts to address this issue by offering a novel perspective on the uncertainty visualisation problem. That is, we are going to use the wealth of established stone to construct the foundations on which we can build a house.

2 Ignoring uncertainty

The comment sentiment expressed by the uncertainty visualisation literature is that we need to visualise uncertainty for the sake of transparency, but what is transparency? Authors become more explicit in what they mean when they directly connect their ideas to the language of statistics by describing it as a degree of confidence (Boukhelifa et al. 2012) or validity (Hullman 2020, Griethe & Schumann 2006) in our conclusions, or as a proportional level of trust in our data that is more detailed than the binary results of a hypothesis test (Correll & Gleicher 2014, Correll et al. 2018). Therefore, we can interpret this to mean that uncertainty visualisation should be acting as a sort of "visual hypothesis test".

A successful uncertainty visualisation should act as a "statistical hedge" for any inference we make using the graphic. Since the purpose of a visualisation is to give a quick gist of the information (Spiegelhalter 2017), this hedging needs to be communicated visually without the need for complicated calculations. Therefore, an effective uncertainty visualisation should not just "show" uncertainty, untrustworthy conclusions should *not be visible*. If we refer to the conclusion we draw from a graphic to be its signal and the variance that makes this signal harder to identify as the "noise", we can summarise the above information into

three key requirements. A good uncertainty visualisation needs to:

1) Reinforce justified signals to encourage confidence in results
2) Hide signals that are overwhelmed by noise to prevent unjustified conclusions
3) Perform tasks 1) and 2) in a way that is proportional to the level of confidence in those conclusions.

Usually visualisations that are unconcerned with uncertainty have no issue showing justified signals, but struggle with the display of unjustified signals. Therefore, we call this approach to uncertainty visualisation as "signal-suppression" since it primarily differentiates itself from from the normal "noiseless" visualisation approach through criteria (2). That is, the main difference between an uncertainty visualisation and a normal visualisation is that an uncertainty visualisation should prevent us from drawing unjustified conclusions.

To look at how we can achieve this goal, it might be helpful to start with some visualisations that are typically considered to not include uncertainty at all so we can understand how they are falling short. Figure 1 shows six different plots, a high and low uncertainty case for three different plots which occur commonly in discussions around uncertainty visualisation. Plots (a) and (b) show the line plot from a simple linear regression on the `mtcars` data, the regression in (a) has been done on the full dataset, and therefore has low uncertainty, while (b) was done on a sample of three points and has very high uncertainty. Plots (c) and (d) show a choropleth map, with county in Iowa being coloured by a simulated measured temperature, with plot (c) depicting a case where the measurement error is low, while plot (d) depictes a case where the measurement error is high. Finally, plots (e) and (f) depict a set of five groups of univariate variables we are looking to compare, where (e) depicts a case where the group variance is high (f) shows a case where the group variance is low. In all plots, we can only choose a single value to represent the distribution, so we opt for the

average of all these distributions. In all plots, the variance in the plots in the top row is significantly different to the plots in the bottom row, so much so, that the trend should actually disappear. Can you see a difference between the plots in the top row versus the bottom row? Is the strength of the trend communicated through the visualisation? The answer to both of these questions is no, the high and low variance cases are identical.

These plots ignore uncertainty and as such, do not perform signal suppression. We will use these plots as a touchstone for our discussion on uncertainty visualisation, and return to them when we discuss the approaches suggested in the literature. At each stage we will assess the approaches based on how well they hide spurious relationships, and reveal the hidden variance structures in the data.

3 Current Approaches

- Why approaches do/dont work
- Review of current approaches to representing uncertainty in plots, as gathered from discipline journals in information visualisation, statistics, environmental science and psychology.
- Most studies involve intrinsic approaches such as colour, transparency, and those that used extrinsic techniques typically used glyphs, error bars, grid based techniques, or contouring Kinkeldey et al. (2014)
- Padilla et al. (2022) two broad categories of uncertainty visualisation techniques, 1) graphical annotation (mean, error bar, hops) 2) map probability of cI to a visual encoding channel 3) hybrid approaches (e.g. contour boxplot, probability density and interval plot)

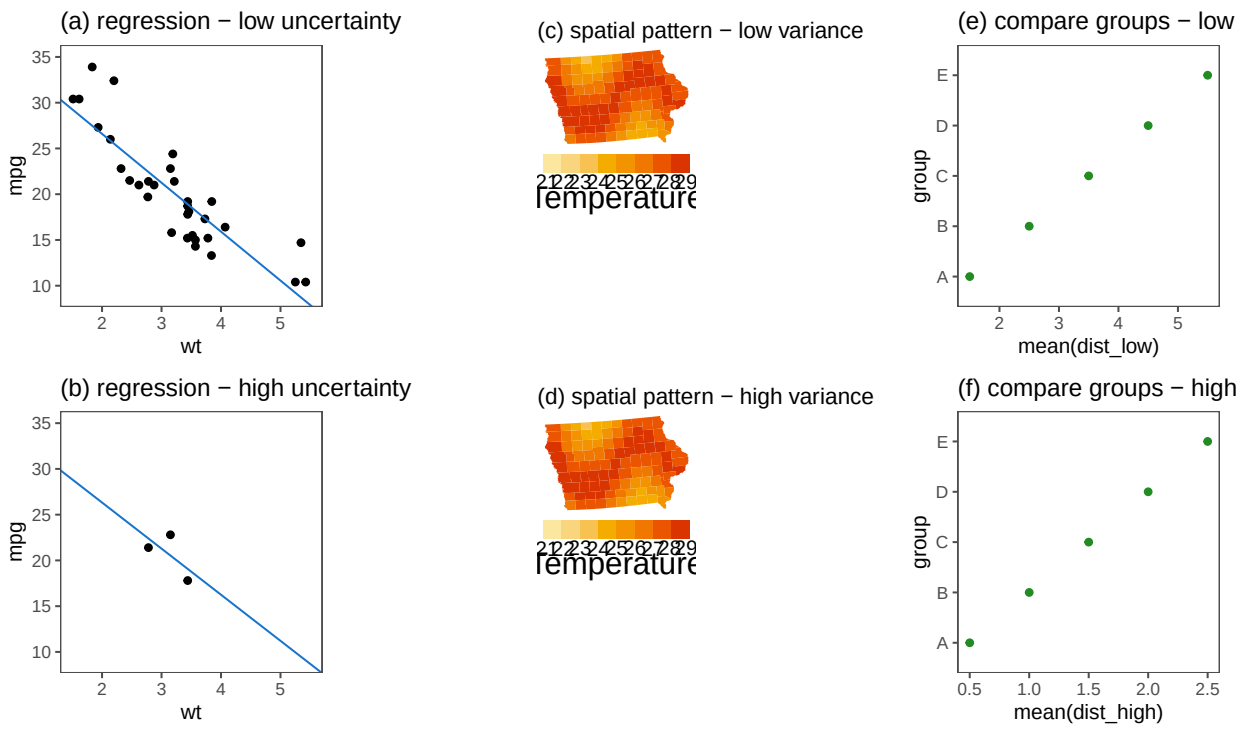

Figure 1: Six different plots, each depicting a high and low uncertainty case for three plots commonly seen in discussions around uncertainty visualisation. The plots are: (a) linear regression with low uncertainty, (b) linear regression with high uncertainty, (c) spatial trend with low uncertainty, (d) spatial trend with high uncertainty, (e) group comparison with low uncertainty, (f) group comparison with high uncertainty. The linear regressions have a downward slope, the spatial map has a sine wave pattern, and the groups show an increasing value in X. These patterns should be visible in the low uncertainty case, but almost invisible in the high uncertainty case. As each plot only shows the central estimate of each uncertainty value, the high and low scenario look identical for each plot.

- Elements from a single distribution should be displayed using a single plot, since displaying the features of one distribution across multiple plots makes the information hard to combine and results in some details (such as the estimate error) being completely ignored [Moritz et al. (2017); Correll2018].

3.1 Uncertainty as a statistic

?@fig-exceed depicts an exceedance probability map that was designed as an alternative to the choropleth map to improve decision making under uncertainty (Kuhnert et al. 2018, Lucchesi et al. 2021). A keen viewer may notice that the exceedance probability map is actually just a choropleth map, only the statistic being displayed has changed. We are not sure it is productive to categorise this visualisation as an uncertainty visualisation.

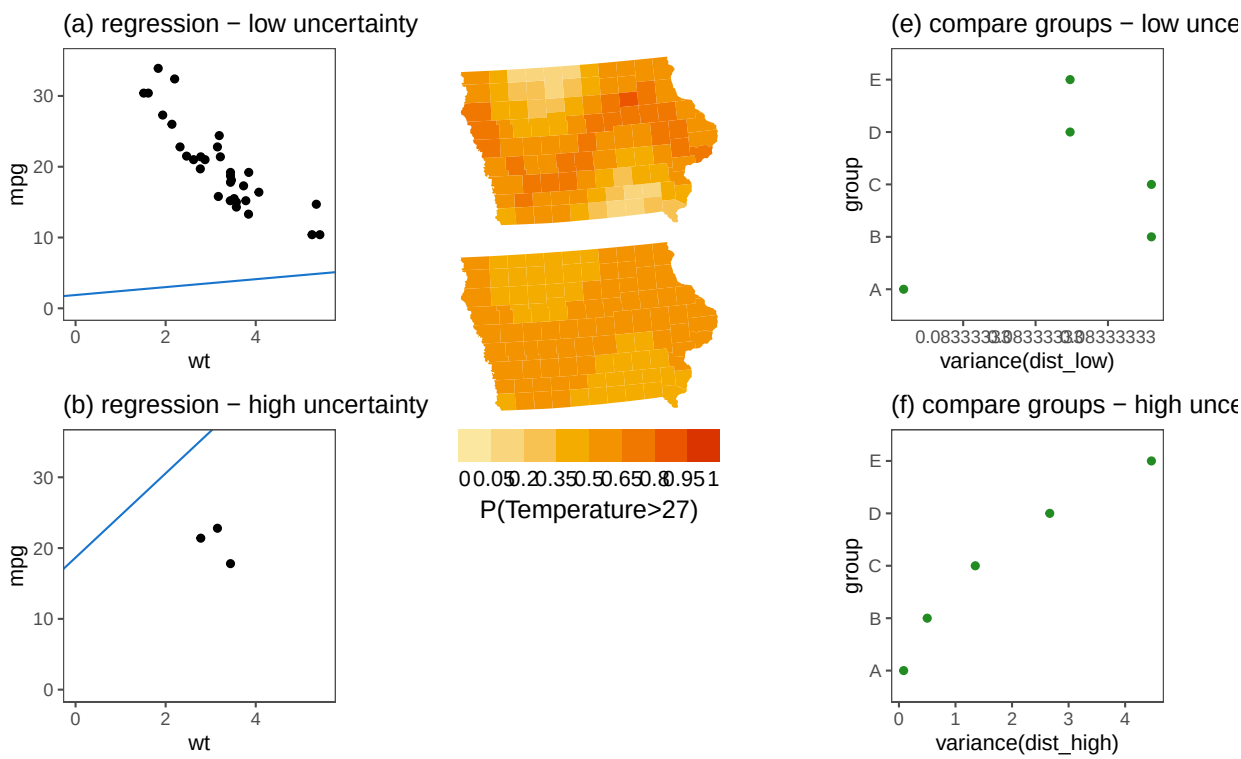

Figure 2: blah

Uncertainty visualisation is not only motivated by signal-suppression, and we would be re-

miss if we did not mention these alternative approaches. Some authors explicitly state it as a variable of importance in of itself (Blenkinsop et al. 2000). While uncertainty can provide useful information, it is important to recognise that the "uncertainty" in these cases is not acting as uncertainty at all. It is acting as signal. This is obvious for the cases where we are explicitly interested in the variance or error, as we are literally trying to draw conclusions about an statistic that is used to describe uncertainty. The uncertainty visualisations that would perform the best in these cases would simply display the uncertainty statistic we are interested in, such as the variance, or probability of an event, using existing visualisation principles. This is precisely what we observe in the literature.

There seem to be two different definitions of uncertainty visualisation floating around in the literature. The first considers *any* visualisation of error, variance, or probability to be an uncertainty visualisation. The second believes an uncertainty visualisation is the output of a function that takes a normal visualisation as an input, and transforms it to include uncertainty information. The former group believe the purpose of uncertainty visualisation to provide signal about a distribution, while the later believe it should act as noise to obfuscate a signal. The lack of explicit distinction between these two motivations leaves the literature muddled and reviewers struggle to understand if uncertainty should be treated as a variable, as metadata, or as something else entirely (Kinkeldey et al. 2014). This disagreement creates constant contradictions in what the literature considers to be an uncertainty visualisation. For example Wilkinson (2005) mentions that popular graphics, such as pie charts and bar charts omit uncertainty, and Wickham & Hofmann (2011) suggests their product plot framework, which includes histograms and bar charts, should be extended to include uncertainty. However, pie charts, bar charts and histograms have all been used in a significant number of uncertainty visualisation experiments (Ibrekk &

Morgan 1987, Olston & Mackinlay 2002, Zhao et al. 2023, Hofmann et al. 2012). If you view an uncertainty visualisation as a function applied to an existing graphic, then you would not see a pie chart or bar chart as uncertainty visualisations. These charts are they are yet to have the uncertainty visualisation function applied to them. If you view an uncertainty visualisation as any graphic that depicts a statistic then there are no limitations on which graphics can or cannot be uncertainty visualisations.

When we use the term uncertainty visualisation to refer to graphics that simply communicate a variance or probability, we are classifying visualisations by the data they display, not their visual features. Graphics, just like statistics, are not defined by their input data. A scatter plot that compares mean and a scatter plot that compares variances are both scatter plots. Given that there is no special class of visualisation for *other* statistics (such the median or maximum) there is no reason to assume visualisations that simply depict a variance, error, or probability to be special.

Of course, there is nothing wrong with explicitly visualising variance, error, bias, or any other statistic used to depict uncertainty as a signal. Just like any other statistic, these metrics provide important and useful information for analysis and decisions. However, there is no interesting visualisation challenge associated with these graphics, and they do not require any special visualisation techniques. The uncertainty in these graphics are acting as a signal variable, and they should be visualised using existing principles of graphic design

- Common uses of the term "uncertainty visualisation"
- Provide some background to the use of the term "uncertainty visualisation"
- Tie it to statistical definitions of uncertainty and concepts of inference

- Wilkinson (2005) also suggested signal supression as the goal of uncertainty visualisation "This chapter is about graphics that guide, qualify, or soften our judgments of uncertain data."
  - Also applies some of the logic occasionally in the direction and stuff, "Some of the letters (B, P, X) almost disappear because they are frequently confused with other letters."
- @[Kay2019] "It is important to have clear, domain- and task-relevant guidance on how to determine how much value to suppress—or if to suppress value at all—due to uncertainty".
- @[geointerviews] On confusion of uncertainty as signal or noise: "Also cases where our experts expressed uncertainty a a source of inspiration for future research."
- Griethe & Schumann (2006) clearly sees uncertainty as additional information or data, and therefore sees the uncertainty visualisation problem as a specific case of the high dimensional visualisation issues.
- Some authors implicitly suggest that visualisations of variance or probability are differentiated due to the psychological heuristics involved in interpreting uncertainty (Hullman et al. 2019). There is no evidence that this psychological effect translates to issues with the visual representation of uncertainty. Again, given that we do not make these same visual considerations for other variables that elicit distaste or irrational behaviour, there is no reason to assume this is what makes uncertainty visualisation so special.

3.2 Uncertainty as a variable

3.2.1 Independent variable

Figure 3 depicts a variation of the choropleth map, where we have a two dimensional colour palette. In this graphic, temperature is still mapped to hue, but the variance is included by utilising colour saturation. While these two maps *do* look visually different (which was not the case in the choropleth map) the spatial trend is still clearly visible in both graphics. This means the uncertainty *is technically* being communicated, however the main message of the graphic is still the spatial trend (that may not exist). The graphic did not suppress the invalid signal, so it is not performing signal-suppression as we would like. At this point, it might be reasonable to ask, why? Why is including the uncertainty as a variable insufficient to achieve signal-suppression, and what changes should we make to ensure signal-suppression occurs?

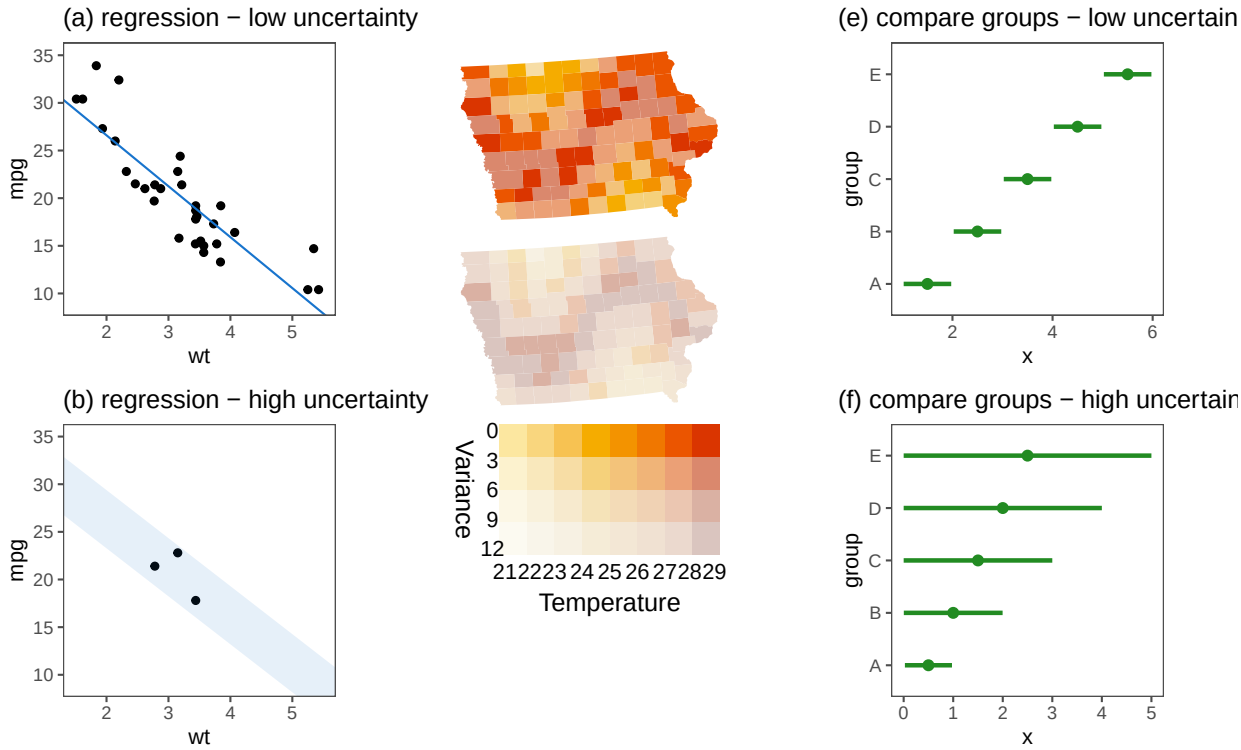

Figure 3: blah

Describing what uncertainty actually is, is surprisingly hard. Most authors simply avoid the problem and describe the characteristics of uncertainty, of which there are plenty.

Specify difference between uncertainty and variance Begg et al. (2014) highlight that uncertainty is related to not knowing a specific value, while variability refers to the range of values a quantity can take at different locations, times or instances. Spiegelhalter (2017) made sure to comment on the difference between precise random events (such as the probability associated with a coin flip), and uncertainty (such as the estimated probability associated with a coin that might be biased).

Often, uncertainty is split using an endless stream of ever changing boundaries, such as: - Whether the uncertainty is due to true randomness or a lack of knowledge (Spiegelhalter 2017, Hullman 2016, Walker et al. 2003) - If the uncertainty is in the attribute, spatial elements, or temporal element of the data (Kinkeldey et al. 2014) - Whether the uncertainty is scientific (e.g. error) or human (e.g. disagreement among parties) (Benjamin & Budescu 2018) - if the uncertainty is random or systematic (Sanyal et al. 2009) - Statistical or bounded (Gschwandtner et al. 2016, Olston & Mackinlay 2002) - Recorded as accuracy or precision (Griethe & Schumann 2006, Benjamin & Budescu 2018) - Which stage of the data analysis pipeline the uncertainty comes from (Walker et al. 2003) - How quantifiable the uncertainty is (Spiegelhalter 2017, Walker et al. 2003)

- Gustafson & Rice (2019) organises uncertainty into the four groups: Deficient: Lack of knowledge; Technical: modelling approximations and measurement error; Scientific Unknown unknowns; Consensus: Disagreement among parties
- Padilla et al. (2022) mention aleatric, epistemic, ontological divide and also differentiates risk with uncertainty
- Mentioned alleatory vs epistemic uncertainty, in the visualisation literature alleatory

uncertainty is more commonly the focus. also mentions dependence and joint pdfs as a consideration Hullman (2016)

- mentions uncertainty can be variance, precision, accuracy, reliability, or related concepts Hullman (2016)
- @[McNutt2020] Have several elements that they consider to cause "visual mirages", from improper imputation of missing values, to summary functions obfuscating the variabiliy of the data. It seems to be a bizzare mismatch of different types of information that are grouped together with "visual mirage".

Kinkeldey et al. (2014) categorised uncertainty according to five criteria depicted in fig-tax which considers if a visualisation is implicit or explicit, intrinsic or extrinsic, visually integrable or seperable, coincident or adjacent, and static or dynamic. A similar version of this taxonomy was presented by Padilla et al. (2022) who commented that visualisation can be organised into two categories, "graphical annotations of distributional properties" and "visual encodings of uncertainty" which seems to functionally align with the intrinsic/extrinsic distinction by Kinkeldey et al. (2014). Griethe & Schumann (2006) organised uncertainty visualisations into two cases (1) a hypothesis test was preformed to confirm the validity of the visualisation and (2) the visualisation has uncertainty depicted. Potter et al. (2012) organised several existing uncertainty visualisations into groups based on the dimensionality of the data (1D, 2D, 3D, and No Dimension) and the dimensionality of the uncertainty (Scalar, Vector, Tensor). However, because the term "PDF", function that is used to describe a random variable, is used to describe both the data and the uncertainty for all dimensions. Grewal et al. (2021) created a taxonomy that mapped uncertainty visualisations to some point in a 2D space defined by the "domain expertise" and "continuum of discreteness" (that scaled from "point estimate" to "continuous distribution"). There

are enough qualitative descriptors of uncertainty to fill a paper, but, a lot of it amounts to rearranging deck chairs on the titanic. The distinctions are often accurate, but have little, if any, impact on how we should visualise, quantify, or discuss the uncertainty.

Rather than trying to define uncertainty by looking at the myriad ways in which it *does* appear in an analysis, we may find it easier to look at where it *does not*. Descriptive statistics describe our sample as it is and summarises large data down into an easy to swallow format. Descriptive statistics are not seen as the primary goal of modern statistics, however, this was not always the case. In 19th century England, *positivism* was the popular philosophical approach to science (positivists included famous statisticians such as Francis Galton and Karl Pearson). Practitioners of the approach believed statistics ended with descriptive statistics as science must be based on actual experience and observations (Otsuka 2023). In order to make statements about population statistics, future values, or new observations we need to perform inference, which requires the assumption of the "uniformity of nature", that is, we need to assume that unobserved phenomena should be similar to observed phenomena (Otsuka 2023). Positivists believed referencing the unobservable was bad science. In other words, these scientists embraced descriptive statistics due to the inherent certainty that came with them. Since uncertainty is non-existent in descriptive statistics, it is clear that uncertainty is a by-product of inference.

This history lesson illustrates what uncertainty actually is. At several stages in a statistical analysis, we will violate the uniformity of nature assumption. Each of these violations will impact the statistic we have calculated and push it further from the population parameter we wish to draw inference on. Uncertainty is the amalgamation of these impacts. If we do not violate the uniformity of nature assumption at any point in our analysis, we do not have any uncertainty.

3.2.2 Combined variable?

Instead of hoping that uncertainty might collapse signal values into a single dimension, we can do some of that work ourselves. As a matter of fact, some uncertainty visualisation authors already have.

There is another benefit to mapping uncertainty to saturation that is not directly related to integrability. As the saturation decreases colours become harder to distinguish. This means high uncertainty values are harder to differentiate than low uncertainty values. We can leverage this implicit feature of colour value by transforming the visual feature space ourselves.

The Value Suppressing Uncertainty Palette (VSUP) (Correll et al. 2018), was designed with the intention of preventing high uncertainty values from being extracted from a map. Since the palette was designed with the extraction of individual values in mind and it has only been tested on simple value extraction tasks (Correll et al. 2018) or search tasks (Ndlovu et al. 2023), it is unclear how effective the method is at suppressing broader insights such as spatial trends.

The methods used by the VSUP bring to light a slight problem with uncertainty visualisation. Specifically, uncertainty and the purpose of visualisation are somewhat at odds with one another. There are two primary motivations behind visualisation, communication and exploratory data analysis (EDA). Communication involves identifying a signal we want to communicate and designing a visualisation that best conveys that, while EDA involves creating a versatile visualisation and using it to extract several signals. If we are designing an uncertainty visualisation for communication then we can just suppress the specific signal we are seeking to communicate. In the map example, we would consider Figure 4 to be a

success as the only signal we are concerned with is the spatial trend.

The lack of versatility of the VSUP is easy to see with a simple example.

Let's say we have a graphic that depicts a set of coefficients from a linear regression and the value of the coefficient is shown using a single colour. We want to know "Which of these coefficients are different from 0?" as well as "Which of these coefficients are different from each other?". To answer this question we do a series of t-tests on these estimates. All of the individual t-tests of fail to reject the null hypothesis that the coefficients are different from 0. We then make a visualisation that suppresses this signal and ensures that all of the estimates are visually indistinguishable from 0. Next, we conduct two sample t-test and find that several of the values need to be visually distinguishable from each other. The VSUP method must pick a single colour for each estimate, and these colours must be *either* visually distinguishable or indistinguishable from each other. We cannot perform signal-suppression on both these signals simultaneously.

This example highlights a fundamental problem with the VSUP that extends to the bivariate map as well. When we blend these colours, we need to decide at what level of uncertainty to blend these colours together. Even though the bivariate map does not explicitly combine colour values at certain variance levels, the mapping of variance to colour saturation does this implicitly. That is, at certain saturation values the colours in a bivariate map are imperceptibly different and appear as though they are mapped to the same value. At this point, it is irrelevant whether or not the colours are technically different, they are the same colour in the human brain. This is of course complicated by the fact that human colour perception varies at an individual level. Some women are believed to have four different cone cells which allows them to perceive a greater range of colours, while others only have one or two types of cone cells and have colour deficiencies (Simunovic 2010)

For VSUP to function for all individuals, we must calibrate each plot to an individual's ability to perceive colour.

If we only use a single colour to express each signal-suppressed statistic, we will always need to decide which signals we suppress and which we do not. This issue has already been raised in the literature. Which hypothesis are suppressed and which are not largely depends on the method used to combining colours in the palette (Kay 2019). The VSUP in Figure 4 used the tree based method that was used by Correll et al. (2018), but there are alternatives that are more appropriate for different hypothesis.

Uncertainty visualisation for EDA would be possible if we designed a plot in such a way that suppressing individual estimates using their variance would naturally extend to broader suppression of plot level insights. This assumption is commonly made by visualisation researchers in normal visualisation experiments (North 2006). If we could express the statistic of a cell using multiple colours, this limitation may disappear entirely. Figure 4 is a visualisation of the Iowa temperature data using a VSUP to colour the counties. The low uncertainty case still has a visible spatial trend, while the spatial trend in the high uncertainty map has functionally disappeared. This means the VSUP has successfully suppressed the spatial trend in the data. However, the spatial trend may not be the only signal of concern in our graphic. Now we must return to the original signal-suppression criteria and ask ourselves if they have all been met. Are all the justified signals reinforced, while all the unjustified signals are suppressed? Is a graphic that performs perfect signal-suppression even possible?

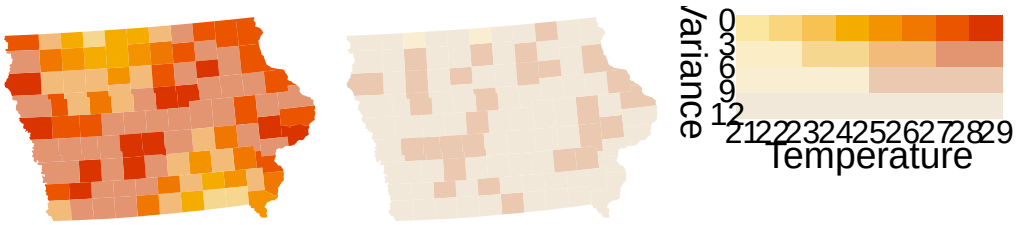

Figure 4: A map made with a VSUP. The counties of Iowa are coloured acording to its average daily temperature and the variance in temperature. Similar to the bivariate map, temperature is mapped to hue while variance is mapped to saturation. Unlike the bivariance map, the colour space we are mapping our variables to has been transformed so that high variance estimates are harder to discern from each other. This map successfully reduces the visibility of the spatial trend in the high uncertainty case while maintaining the visibility of the spatial trend in the low uncertainty case.

3.3 Uncertainty as a random variable

Figure 5 displays a pixel map (Lucchesi et al. 2021), which is a variation of the choropleth map where each area is divided up into several smaller areas, each coloured using outcomes from the larger area's (i.e. the county's) average temperature sampling distribution. The spatial trend is clearly visible in the low variance case, but functionally disappears in the low variance case. While the spatial trend is just barely visible in the high uncertainty case, it is much harder to see. This means the graphic also achieves the third criteria for signal-suppression, i.e. our difficulty in seeing the distribution is proportional to the level of uncertainty in the graphic.

3.3.1 EDA thing?

- We will not go through how this uncertainty is quantified, but we will rely on the fact that all uncertainty, no matter how abstract, can always be quantified by a random

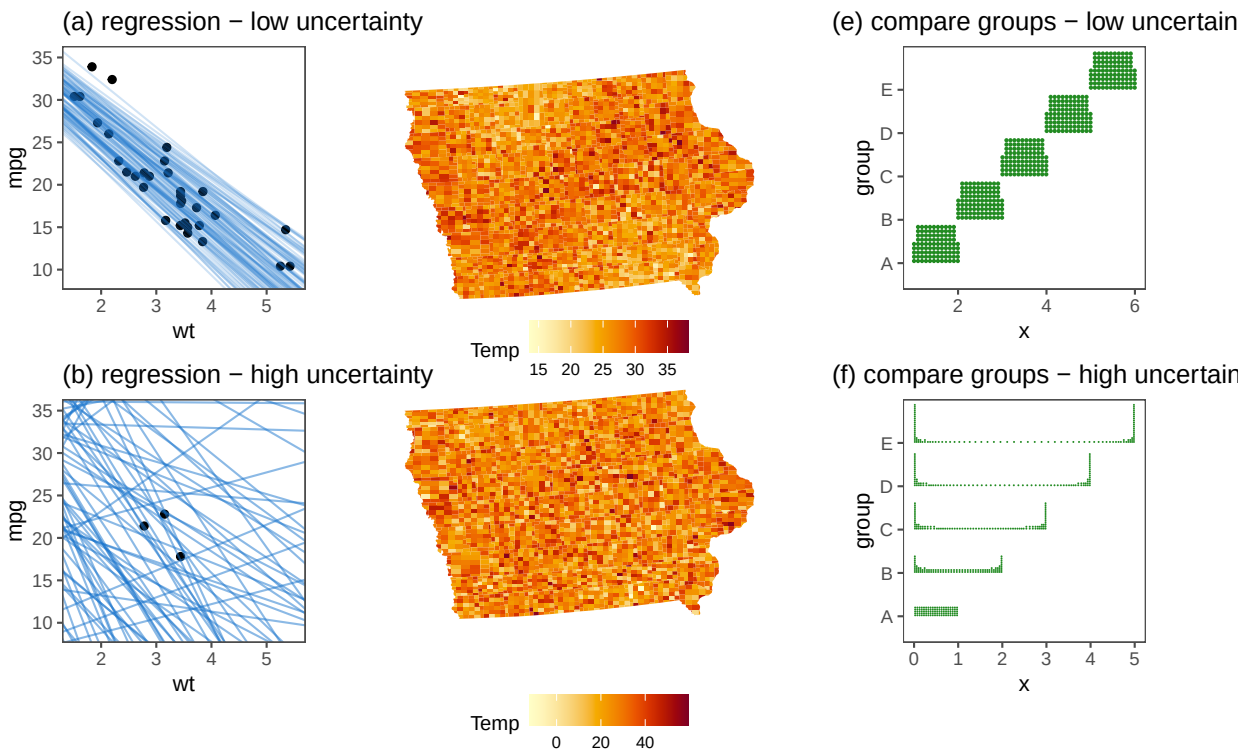

Figure 5: blah

variable

- We can see how this works with a basic measurement error example. If we record our daily maximum temperature as 23 degrees celsius on a device with +-1 degree of error, then we can represent the observation as a uniform $U(22, 24)$ random variable *assuming we want to represent the true temperature*. If we want to represent the recording itself, it is 23 degrees celsius, no uncertainty needed. It is the act of inference that creates uncertainty, and the uncertainty created can be represented by a distribution.

However, it is not uncommon for authors to express a desire for uncertainty visualisations that perform signal-suppression in visualisations made for EDA (Sarma et al. 2024, Griethe & Schumann 2006). For uncertainty visualisation for EDA to work, we would need to assume that suppressing individual estimates using their variance should naturally extend

to broader suppression of plot level insights.

Earlier we established that uncertainty is a by-product of inference, which means without inference, there is no uncertainty. Often EDA is used to give us an understanding of our data and identify which signals are worth pursuing. In this sense, EDA is the visual parallel to descriptive statistics, as it is performed without an explicit hypothesis, which means there is no inference, and by extension, there is no uncertainty.

Some authors recognise inference will always occur (in some shape or form) and believe uncertainty *should* be visualised but do not recognise *how* uncertainty would be visualised. Hullman & Gelman (2021) argued that there is no such thing as a "model-free" visualisation, therefore all visualisations require uncertainty as we are always performing inference. While it is true that we can think of visualisations as containing implicit inferential properties, there are many potential inferences in any single visualisation. This makes it a little difficult to ensure uncertainty is always included. For example, if we have a visualisation that shows an average, we would need to identify if the signal suppression should be performed using the sampling variance or the sample variance (Hofman et al. 2020). The distribution we use depends on the inferential statistic but until the viewer chooses one and thinks about it (which isn't easily observable), the particular variety of uncertainty which would need to be displayed can't be calculated. This means the ideal uncertainty visualisation should not only meet the signal suppression requirements, but should also endeavor to be versatile enough to meet those requirements for all the signals displayed in the graphic.

3.3.2 Quantify all of it?

quantification - In order to visualise uncertainty, it needs to be quantifiable [Griethe & Schumann (2006); Leland2005], in order to quantify uncertainty, it needs to be mathemati-

cally defined. The uncertainty visualisation literature is completely awash with papers that define uncertainty using a vague encompassing phrase, (e.g. Walker et al. (2003) defined it as any deviation from complete determinism) but go on to quantify uncertainty as a PDF, error, or some other easily quantified mathematical object when it comes time to do the visualisation. This definition swap happens so subtly that it seems to go unnoticed by authors

- Risk communication also needs to have clear objectives, use plain language, limit information to only what is necessary, and segment the audience to allow for differences in interest and knowledge (Spiegelhalter 2017).
- Padilla et al. (2022) do specify that they are going to refer to quantified uncertainty that can be visualised ("most commonly as a probability distribution")
- Griethe & Schumann (2006) mentions that there are detailled classifications of uncertainty but the literature and subsequent techniques are not ballanced, and instead concentrate on scalar values. In facing this lack of ballance she creates a new classification system for uncertainty visualisation.
- Probability judgements are less relative than other psychophysical judgements, 0%, 50% and 100% all have well understood meanings. Subjective probability distributions are spontaneous, and subjects are constructing a distribution upon being asked rather than articulating a fully formed mental representation of their beliefs. Therefore it is subject to heuristics. consider anchoring a scale with farmiliar probabilities . not sure if being asked for the entire distribution helps, e.g. the graphical one reduces noise but asking people for an interval estimation leads to overprecision and overconfidence Hullman (2016)
- Even something as simple as sketching a distribution before recalling statistics or

making predictions can greatly increase the accuracy of those measures (Hullman et al. 2018, Goldstein & Rothschild 2014).

pipelines - Griethe & Schumann (2006) commented that uncertainty appears at every step in the analyis pipeline, and commented on some the mathematical definitions that seek to quantify uncertianty, but did not explain how these mathematical definitions can be combined across levels. Unfortunately this quantification is needed since she mentioned that in order to visualise uncertainty, it needs to be measurable. - Fischhoff & Davis (2014) takes this approach by considerting how decisions with different levels of flexibility should have different uncertainty visualisations. If the user has no choice they should simply be made aware of the risk, if they have several options, the user needs to see the uncertainty in each, if they need to come up with the options themselves, they need to see how things work and how the system fails. - Padilla et al. (2022) "can enter the analysis pipeline during the measurement, modeling, and forecasting phases"

These divisions create Thomson et al. (2005) discussed the problem of multiple times of uncertainty and suggested mapping each type of uncertainty (such as accuracy, precision, credibility, etc) separetely and then have the analyst combine them in their head. To map each of these variables separately suggests that each level of uncertainty is somehow its own variable. While this is possible, and relevant if

If you are already confused by this distinction, it is about to get worse for you, because these can technically be performed at the same time. This can occur is a scenario that covers many cases that researchers will consider to be too uncertain to quantify, something called "deep uncertainty" by Hadjimichael et al. (2024). These abstract uncertainties can include abstract concepts like expert disagreement, the credibility of a source (Thomson et al. 2005), forecaster confidence (Padilla et al. 2021), or alternative decisions that could

have been made in the analysis (Kale et al. 2019) (something that frequently see visualised in climate scenario uncertainty). If the authors are willing to quantify them, these cases can almost always be covered by distributions of distributions, and can lead to a `ggdibbler` version of a `ggdist` plot. Note that we cannot have a `ggdist` version of a `ggdibbler` chart.Lets look at two examples of this to see what we mean.

We will admit that cases where the uncertainty is not published, as discussed by Peña-Araya et al. (2025) are a little more difficult, but still possible. If something is worth doing, it is worth doing badly, so imputing the distribution of random variables using the resources available is better than not ignoring it entirely.

case of "deep uncertainty"

This occurs when you have a distribution of distributions. These cases are often thought to be "unquantifiable" by other researchers, but that is not entirely true, they can be represented by random variables.

- (Thomson et al. 2005) Analysts are also concerned with abstract uncertainty such as credibility of a source, and as uncertainty becomes more abstract, it is more difficult to quantify, represent and understand.
- (Hadjimichael et al. 2024) Say that uncertainty can't be quantified by a probability distribution because experts do not agree and call it "deep uncertainty"
- (Peña-Araya et al. 2025) In interviews with geologists expressed they had to "work with other sources of uncertainty that were more abstract and less easy to pinpoint or measure"
- (Peña-Araya et al. 2025) Uncertainty metrics are not systematically published along tephra studies (from geologist interview)

- e.g. climate change scenario and forecast uncertainty

While each random variable level can only hold one role, These goals cannot be performed simultaneously on the same distribution, but they *can* be performed on the same plot. We can theoretically have a `ggdibbler` version of a `ggdist` plot.

Now, a list of examples is helpful, but not exactly sufficient to describe the greater problem we are facing here.

3.3.3 Statistical inference and exploratory data analysis

- @[Hullman2021] "Researchers have commented on uncertainty as implicit in exploratory and interactive analysis"
- @[Hullman2021] Issues with EDA e.g. checking multiple hypothesis which makes them less valid (more likely to find spurious relationships) - I think of EDA as a search for hypothesis rather than a confirmation of any hypothesis.

Creating a concrete and well-defined uncertainty visualisation system has proven difficult due to uncertainty's abstract and omnipresent nature. This is made worse by the fact that uncertainty often plays two roles in our analysis: one of noise and one of signal (Mason et al. 2024). Uncertainty as a topic of inference in of itself is uncertainty as signal, while the uncertainty that *interferes* with our inference is uncertainty as noise. We argue that uncertainty acting as noise is the only uncertainty we should be concerned with, but the conflation of the two approaches creates a lot of confusion in the literature. This confusion is not only created by the "uncertainty" component of uncertainty visualisation, but it can also be created by the dual roles of visualisation itself.

Visualisations can simultaneously be used for exploratory data analysis (EDA), which is the process of searching for interesting hypotheses, and for confirmatory data analysis (CDA),

the process of verifying a hypothesis (**?**). Some have argued that the two methods are not entirely distinct, as it is often difficult to draw a hard line between EDA and CDA (Hullman & Gelman 2021). We would argue that the approaches exist on a continuum, where we have an inherent trade-off between the number of hypotheses we can look for and the certainty of any conclusions reached. It can help to think of the knowledge-generating process of EDA and CDA as the settings on a hose, where EDA is a fine misting spray that touches everything in the room, and CDA is a high-pressure jet capable of obliterating any and all debris from any single spot.

The quest for a flexible uncertainty visualisation tool is fundamentally a quest for a tool that can perform EDA. The problem with this goal is that EDA has uncertainty that is inherent to the approach, which is often conflated with the uncertainty that comes from our data. The EDA versus CDA dichotomy is better known as inductive versus deductive reasoning within formal logic, where uncertainty in our conclusions is inherent to induction, but not deduction. This means that EDA, inductions counterpart, has uncertainty in any conclusions reached through the method, with the requirement to follow up our new-found hypothesis with CDA if we want true certainty. Visualisation authors often try to account for this inherent uncertainty, believing the uncertainty inherent to EDA can somehow be incorporated into our plots. This causes authors to try to perform some kind of statistical super-test, where they pre-emptively hedge against every false inference that could possibly be drawn from a graphic (**?**Hullman & Gelman 2021, **?**). This is typically done by visualising a null distribution alongside the observed data on a single plot, an approach that is statistically identical to the line-up protocol (Buja et al. 2009). These methods are, fundamentally, deductive reasoning, and therefore not EDA. This is not to say these approaches are pointless; the line-up protocol is a particularly useful alternative

to hypothesis testing, as they allow for statistical inference when there is no classical test available (Hofmann et al. 2012). Rather, the problem is just that it is not EDA. The sum total of a lot of CDA is not EDA, just as frantically swinging a high-pressure jet around a room is not equivalent to using a misting spray. Even if you do manage to wet everything, you are going to use an exorbitant amount of water to do it (and possibly also break a few personal items). To truly create an uncertainty visualisation approach that is capable of EDA, we need to accept that uncertainty is inherent to the method, and it cannot be pre-emptively removed from the visualisations. If we accept this fact, then the only possible source of uncertainty in our "flexible uncertainty visualisation system" is from the data itself.

Viewing the input to an uncertainty visualisation as a distribution is hardly new territory. Kay (2023) originally suggested thinking about uncertainty visualisations with distribution inputs when developing `ggdist`. He proposed a distribution input to replace the commonly used mean and standard deviation, as the latter requires us to assume a Gaussian distribution. We would even take this point a step further and assert that a distribution is sufficient to characterise *any* uncertainty, no matter how intangible it is. This is in contrast to the assertions made by some authors, who believe abstract notions of uncertainty, such as credibility (Thomson et al. 2005), forecaster confidence (Padilla et al. 2021), or uncertainty about uncertainty (Hadjimichael et al. 2024), are too complex to be quantified. This is simply untrue, as abstract concepts such as human belief are regularly quantified by Bayesians and hierarchical models are often used for uncertainty about uncertainty. As a matter of fact, since uncertainty is always a by-product of statistical inference, it should always be possible to quantify it using a distribution, with the only distinction between "quantified" and "unquantified" being an inferential statistic and willingness to quantify

the uncertainty (Mason et al. 2024).

Quantifying uncertainty and representing it in the data is fundamentally part of the data manipulation stage of our analysis. Therefore, according to the philosophy proposed by Wickham (2010), it should be kept as separate as possible from the visualisation stage of our analysis to give users full transparency in *what* precisely they are visualising (Wickham 2010). In `ggdibbler`, this separation is created using `distributional` (O'Hara-Wild et al. 2024), which allows users to store distributions inside data frames as individual cells. We will work from the assumption that users have already quantified the uncertainty as distributions before attempting to visualise it, and simply design a system that allows for `distributional` inputs. If readers are looking for help on how to do this, you may be better off reading Mason et al. (2024), as that will not be the focus of this work. Instead, we will discuss how to redesign a data visualisation such that we can pass a random matrix just as easily as we can pass our deterministic data frame.

3.3.4 Show me the data

- we should aim to show enough information to solve a task while avoiding irrelevant distracting information (Kosslyn 2006).
- (Unwin 2020) "One persons statistics might be another person's raw data"

To thoroughly understand why pixel maps seem to successfully perform signal suppression, we would need a more thorough discussion than what can be offered here. However, this does not prohibit us from discussing potential reasons for the graph's effectiveness. There is a second type of uncertainty that the pixel map quietly communicates but we have not touched on yet, and this uncertainty is also conveyed by visualisations of our raw data. That is, these graphics can show assumption violations.

As we discussed in previous sections, we can consider uncertainty to be "the amalgamations of the impacts of violations to the assumption of the uniformity of nature". It's not a definition that rolls off the tongue, but we can work with it. Thankfully, this definition of uncertainty aligns nicely with all the concepts that are included in the uncertainty umbrella. Some works (Hullman et al. 2018, Maceachren et al. 2012, Thomson et al. 2005) focus narrowly on specific terms with mathematical definitions, such as probability, confidence intervals, variance, error, or precision. These works are only concerned with quantifying the final impact of uncertainty on our statistics. That is, how large should the bound around our statistic be, such that our "true" statistic can be inferred. Others (Griethe & Schumann 2006, Wilkinson 2005, Pang et al. 1997, Pham et al. 2009, Boukhelifa et al. 2017) include broader loosely related elements, such as missing values, reliability, model validity, or source integrity. These broader and harder to quantify concepts are concerned about potential sources of uncertainty, that is, they describe violations to the assumption of the uniformity of nature.

What this means is that we have two types of uncertainty, but one is more "processed" than the other. Quantifiable uncertainties are just assumption violations expressed as an effect on our final statistic (or true variation of our data). The disconnect between quantifiable and unquantifiable uncertainty creates a huge problem for authors trying to visualise it. A survey of visualisation authors cited "not knowing how to calculate uncertainty" as one of the primary reasons they did not include it in visualisations (Hullman 2020).

There are two reasons we might leave our uncertainty as an assumption violation rather quantifying the effect. The first reason is that we may be unable to translate the assumption violation to a quantifiable uncertainty. There is no blanket rule that allows us to reliably quantify all uncertainty for every statistic, although some researchers have tangled with the

idea. For example, Thomson et al. (2005) suggests a mathematical formula for *examples* of uncertainty, and information theory tries to quantify uncertainty using the idea of entropy, but they ignore the disconnect between the broad concept of uncertainty and what we can reliably quantify. Some authors don't believe that it is even possible to quantify all the assumption violations (Spiegelhalter 2017).

The second reason to leave uncertainty as a potential violation of our assumptions, is that we might not know the final statistic we are seeking to calculate. This is the case for visualisations made for EDA, and a large number of developments in EDA visualisation have been in displaying these difficult to quantify violations. For example, Tierney & Cook (2023) builds upon the tidy data principles to allow users to handle missing values. This includes data plots with a missing value "shadow" that allows visualisation authors to identify if the variables used in a plot have any structure in their missing values, which would contribute to uncertainty.

With this understanding it becomes clear to see why uncertainty is tied to an endless string of examples in the data analysis pipeline. Uncertainty examples include imputed data, model selection, inherent randomness, biased sampling, etc., not because these things *are* uncertainty, but because they *create* uncertainty when we perform inference. Whether or not these elements are relevant is highly dependent on what statistic you are trying to draw inference on, and by extension, the purpose of your visualisation.

This relationship between uncertainty and the purpose of our analysis is littered throughout the literature. - Mention the terminology shift, where infoviz people seem to refer to the fact that graphics are related to statistical inference as "task based" - @[geointerviews] Uncertainty is context-dependent, interconnected, and malleable. Whether uncertainty is a source of doubt or not depends on context. - Spiegelhalter (2017) noted we "cannot assess

the quality of risk communication unless the objectives are clear".

Multiple authors have commented on the need to consider quantifying and expressing uncertainty at every stage of a project as the goal shapes every step of the analysis (Kinkeldey et al. 2014, Hullman 2016, Refsgaard et al. 2007). Otsuka (2023) suggested that the process of observing data to perform statistics is largely dependent on our goals, because the process of boiling real world entities down into probabilistic objects (or "probabilistic kind" as he puts it) depends on the relationship we seek to identify with our data. Meng (2014) commented what is kept as data and what is tossed away is determined by the motivation of an analysis and what was previously noise can be shown to become signal depending on the question we seek to answer.

Wallsten et al. (1997) argue that the best method for evaluating or combining subjective probabilities depends on the uncertainty the decision maker wants to represent and why it matters. Fischhoff & Davis (2014) looks at uncertainty visualisation for decision making decides that we should have different ways of communicating uncertainty based off what the user is supposed to do with it.

This makes it very difficult to move quantified uncertainty through the layers of our analysis, especially when designing a visualisation for EDA. If we don't know what the final statistic is, we cannot quantify the effects of our assumptions. Therefore, it is often the case that the best uncertainty visualisation is not an uncertainty visualisation at all, but simply the most accurate depiction of our raw data. This does not mean that visualising raw data will always prevent insignificant signal from getting through. Roy Chowdhury et al. (2015) illustrated how groups that appear linearly separable in a linear discriminant analysis (LDA) visualisation of the data can actually be the result of a LDA performed on too many variables, something that was not clear from the visualisation until the line-up protocol

was implemented. However, just showing the data is a simple but effective option for uncertainty visualisation that seems to be largely overlooked.

Of course, visualising the raw data is not always possible nor suitable. For example, if we have a specific statistic in mind, it may be more appropriate to show the sampling distribution of that statistic rather than the variability of all observations. If we visualise that sampling distribution in a way that allows us to see it's shape, and therefore recognise any assumption violations (such as a non-normal distribution), we maintain some of the benefits that came with visualising the raw data. This is exactly what the pixel map does, and likely contributes to it's success as a signal suppression method.

3.3.5 Show me the data

- rarely used extension is aggregation of uncertainty over an area (retrieve overall estimation from a spatial distribution of uncertainty). Other tasks include comparisons, rankings. also have a task called "search" where participants have to identify entities that fulfill certainty characteristics (high or low values) Kinkeldey et al. (2014)

Rather than trying to figure out how to combine signal and uncertainty into a single colour, we can just display a sample instead and allow the viewer to extract *both* the estimate and the variance.

It is clear that the pixel-map is not only suppressing the false information, but it is doing so by providing *more* information. The efficacy of this method means that visualisations of simulated samples pop up repeatedly in the literature, with examples including samples that are animated over time (Hullman et al. 2015, Blenkinsop et al. 2000), pixel-maps, and spaghetti time series plots. Not only does this method help readers understand the plot level gist, it is also unlikely to damage the viewers ability to extract individual estimates.

Extracting global statistics, such as the mean or variance, from a sample can be done with relative ease, especially when those values are mapped to colour (Franconeri 2021). Additionally, Ndlovu et al. (2023) found that participants applied the same methods they used for simple choropleth maps to complicated uncertainty maps even if that take away was invalid. The pixel map can be read the same way as a choropleth map, which allows us to leverage that automatic response. Therefore, the pixel map performs signal suppression, without sacrificing the viewers ability to extract general statistics, unless those statistics *should* be harder to extract due to the uncertainty in the value. So, what is it about Figure 5 that allows it to be such a successful uncertainty visualisation? Is this a question we can easily answer?

4 Evaluating uncertainty visualisations

If we want to make conclusions about how effective any uncertainty visualisation method is we need to look at the results of evaluation experiments. Unfortunately, despite the abundance of these studies, existing literature reviews have struggled to synthesise them into any common rules (MacEachren et al. 2005, Kinkeldey et al. 2014, **?**, Hullman 2016). This issue is particular to uncertainty visualisation, and pervasive, so it begs the question, what is causing it?

There are as many explanations for the noisy evaluation studies, as there are contradictions in the research itself. Kim et al. (2019) believes there is some interference from participants prior beliefs; Hullman (2016) believes the noise in the literature could come from visual heuristics, subjective probabilities, unknown participant utility functions, or misunderstanding of statistical concepts (such as confidence intervals). Kinkeldey et al. (2014) suggest the best visualisation changes depending on the audience, so we cannot ex-

pect the same results between different subpopulations. Brennen & Tuerk (2018) attributes issues to cognitive load from complicated uncertainty visualisations, and the participants prior experience in a topic, suggesting it is aleviated by measuring confidence and cognative load. While these issues will certainly have some impact on our ability to create a cohesive theory, most of them are not unique to uncertainty, visualisation and fail to explain the problems in this field in particular. Rather, the issue is likely coming from a disconnection between the evaluation method and wider goal of the field.

4.1 Current evaluation methods

The inconsistency in the motivations for uncertainty visualisation bleed into the evaluation of the methods. Treating the results of these papers, as though they are somehow agnostic towards their stated purpose, leads to regularly conflicting results in the field. This problem is pervasive within standard visualisation evaluation literature, where studies are regularly criticised for failing to correctly align questions and plot types (Vanderplas et al. 2020) as though the goals of our analysis and the plots we use are somehow independent. This becomes particularly egregious in uncertainty visualisation because, unlike other statistics, uncertainty plays two, opposite, roles in our analysis. The role of signal is to be visible, while the role of noise is to obfuscate, so flipping the role of uncertainty also flips our conclusions about a methods effectiveness. This problem was also noticed by Brennen & Tuerk (2018), however their solution of measuring cognitive load and confidence would not alleviate this deeper problem. This issue is further aggravated by the fact that authors often provide little justification for their chosen response models (Hullman 2016), and the fact that visualisations are compared on the basis of being "uncertainty visualisations" (Ibrekk & Morgan 1987, Hullman et al. 2015, Hofman et al. 2020), a classification that holds little

to no meaning. This disregard for the wider motivation of the field leads to findings that are confusing and conflicting, which cannot be coalesced into a cohesive theory.

Uncertainty visualisations are most commonly evaluated based on how accurately viewers can extract an estimate and its variance (Kinkeldey et al. 2014, Hullman et al. 2019). This is not unusual, as direct observation is the simplest way to verify that information can be accurately read from a graph (Vanderplas et al. 2020). Unfortunately, this approach doesn't work for uncertainty visualisation. The second we ask a specific question about a statistic, that statistic becomes our inferential statistic, even if the plot was not the intent behind the plot. By shifting the focus from $\hat{X}$ to $Var(\hat{X})$ or $P(\hat{X} < x)$ we end up evaluating visualisations on their ability to convey uncertainty statistics, rather than their ability to perform signal suppression. There is nothing unique to uncertainty in these studies, so when we boil the findings down to generalised results, they simply restate existing principles within information visualisation, even if they do not realise it themselves. Some of the findings are particularly obvious, such studies that found participants were more accurate when reading a probability expressed as a text than when they had to extract if from a graphic (Cheong et al. 2016, **?**). Other studies replicate existing research, such as the finding that a probability mapped to a position is more accurate than one mapped to area (Ibrekk & Morgan 1987, Gschwandtner et al. 2016), a finding that follows the hierarchy of perceptual tasks, which has been established for over 40 years (Cleveland & McGill 1984). This extends beyond simple accuracy evaluations, as we found that colour was more effective than size when searching for maximums and minimums in variances (Sanyal et al. 2009), even though we have known pre-attentive aesthetics such as colour are more efficient for search tasks since the 1980s (Vanderplas et al. 2020). By classifying these studies as evaluations of "uncertainty visualisation" while evaluating uncertainty as signal, we are

encouraged to see successful examples of signal-suppression as failure. This approach leads authors advise against particular aesthetic mappings for uncertainty, because it leads to participants have more difficulty extracting values (Blenkinsop et al. 2000). This conclusion is antithetical to the goals of signal-suppression and occurs because these methods evaluate uncertainty as a signal, not as noise.

If we cannot directly measure uncertainty for fear that it turns into signal, we might then assume we can measure the secondary benefits of increased transparency. This seems to be the approach of many visualisation authors, as secondary benefits such as trust, confidence, and risk aversion are all frequently used in uncertainty evaluation studies (Hullman et al. 2019). Unfortunately, measuring these secondary effects often leads to confusing conclusions that simultaneously argue for, and against, the inclusion of uncertainty. This is most commonly noticed in the use of trust as a measure, as several authors have commented that measuring trust, and not transparency, can lead to a questionable subtext that argues against transparency (Spiegelhalter 2017, O'Neill 2018). Confidence also has the confounding factor that there are many reasons for decreased confidence, including a poorly constructed visualisation, and using it as an evaluation metric collapses these sources down into a single metric. This subtext pops up in the uncertainty visualisation literature where visualisation authors will opt to leave out uncertainty information because it decreases confidence in the conclusions (Hullman 2020). A sentiment that carries over into evaluations of uncertainty visualisation, where Blenkinsop et al. (2000) commented that visually integrable depictions of uncertainty should be avoided, as they decrease the viewers confidence in their extracted data values. Another secondary effect that is similar to trust and confidence is risk aversion, which is an economics term used to describe an agent who would chose a random variable with a lower expected payout because it also has a lower

variance. Risk aversion's role in the uncertainty visualisation is, to be frank, confusing, as it is unclear whether we shoud want more rational decisions that ignore uncertainty, or less rational decisions that incorporate uncertainty. The uncertainty visualisation literature does not clear this up as authors will argue that more risk aversion is better in one paper (Hullman et al. 2019), and argue for less risk aversion (by proxy of suggesting rational agents as a benchmark) in the next (Wu et al. 2023). Ultimately, these approaches have similar issues to the value extraction studies, except they are slightly more confusing in their goals leading them to simultaneously argue for an against the inclusion of uncertainty in a visualisation.

This is made even more complicated by the fact that there is no standard technique to evaluate uncertainty as noise, and attempts to do so are often ad hoc with varying levels of success. Some authors try to incorporate including uncertainty by asking vague questions but compare the results to a ground truth, creating a situation where plots are *evaluated* as a value extraction study without participants being asked the *questions* of a value extraction task. Examples include asking participants for the "best estimate" (Ibrekk & Morgan 1987), or identify when a distribution is "furthest to the right" (Hofmann et al. 2012) and then compare these results to the distributions expected value. The convoluted questions designed to incorporate uncertainty information not only lead to noisy results, but frequently confuse the participants and lead to obviously wrong responses. For example, Hullman et al. (2015) asked participants to compare two groups, A and B, and had many participants say that group A was more likely to be bigger, despite group B having a higher mean. In another study Gschwandtner et al. (2016) ask participants the "probability that the interval has already ended at the marked point in time?" and participants replied with the probability that the interval had already started. The confusion from trying to

incorporate uncertainty uncertainty can extend to the authors of a study, who often tie themselves in knots trying to come up with measurements for the effects of noise. We can see an example of this in the two Padilla et al. (2017) where the first study required participants to assume that an oil rig being "more likely to be hit" by a hurricane would *not* translate to the rig sustaining "more damage", while the second required participants to assume the opposite. This clearly contradictory assumption in how to appropriately use uncertainty information seemed to go unnoticed by the authors. While these approaches are certainly a step in the right direction, the experiments overcomplicate the problem and produce results that are not even consistent within the individual experiment.

This is not to say all all evaluation studies fail to properly evaluate uncertainty as noise. There are several studies that ask participants to identify a particular signal that the noise is trying to obfuscate (Kale et al. 2018, Correll & Gleicher 2014) which seems to be an effective method. The only problem with these studies is that most uncertainty visualisation methods are not integrated into *the grammar of graphics* (Wilkinson 2005, Wickham 2010) so we regularly see comparisons between disparate plots that that would never be considered substitutes for one another outside the artificial "uncertainty visualisation" framework they are placed within. For example Kale et al. (2018) compared static bar charts with error bars to a bar chart with animated samples, meaning that any difference in participants ability to read the plot could be due to the statistic (confidence interval versus sample), the geometry (bars versus intervals), or the use of animation (static versuss animated plots). This means that even when evaluations *are* done correctly, there is no generalisable theory we can take from the results.

The point here is not to mock the existing literature for poor academic merit. If you were to go read these papers yourself, you would find them to (usually) be logically consistent

and well formulated pieces of work. Rather, the point is to illustrate that evaluating uncertainty as a noise is surprisingly difficult. Designing tests for signal suppression will require a formalisation of uncertainty within the grammar of graphics, as well as improved evaluation methods.

4.2 Implicit Hypotehsis Testing

The main problems with current uncertainty visualisation evaluations is that they often require explicit (or convoluted) questions about the variance. Asking direct questions about the statistics or outcomes is not an explicit requirement of visualisation evaluations. In their review of testing statistical graphics, Vanderplas et al. (2020) drew a distinction between explicit tests, where participants are asked direct questions about specific features of a plot, and implicit testing, where users identify both the purpose and function of the plot. The lineup protocol is the most salient example of the implicit approach. Lineups are a confirmatory visualisation tool where participants are shown a set of $M - 1$ plots, and asked to identify the plot that is the "most different", leaving participants to decide what "most different" means to them, even if it is not what the authors intended (VanderPlas & Hofmann n.d.). The implicit testing does not limit the versaility of the approach, with the lineup being used to evaluate the effectiveness of different types of plots (Hofmann et al. 2012), colour palettes (Reda & Szafir 2021), and design decisions (VanderPlas & Hofmann n.d.).

Lineup protocols are not only a useful for their implicit testing, but they also have parallels to hypothesis testing that can be leveraged for uncertainty visualisation. The concept of signal suppression is, at its core, an assertion of statistical validity, which means the visibility of signals should be directly proportional to p-values or some equivalent measure.

This comparison is not new in uncertainty visualisation where parallels have been drawn to frequentist statistics by Correll & Gleicher (2014) who compared results to Cohen's D, and to Bayesian statistics by Kim et al. (2019) who evaluated plots based on their impact on users prior beliefs. The comparison to hypothesis testing is far more natural for the lineup protocol, with **?** explicitly defines a visual test statistic, while Li et al. (2024) compared standard statistical tests to the lineup-protocol using power curves. The parallels to lineups and uncertainty visualisation are numerous, with the relationship even being identified by Hullman et al. (2015) in their development of hypothetical outcome plots.

The lineup protocol and uncertainty visualisation share a lot in common, with the motivation behind lineups also rooted in checking if perceived patterns are real or merely the result of chance (Buja et al. 2009, Wickham et al. 2010). As both approaches are attempting to do the same thing, it is likely that we are unable to leverage the lineup protocol directly to evaluate uncertainty visualisation, but new evaluation methodology should try to learn from the success of the lineup approach. Designing an implicit testing method for uncertainty visualisation that allows us to draw parallels to standard notions of statistical significance, would solve many of the issues with the current evaluation approaches.

4.3 Heuristics research

There is one element of uncertainty visualisation research that we have not touched on yet, but is incredibly consistent, and that is the research around visual heuristics. The uncertainty visualisation heuristics research, unlike the plot evaluation work, is increbidly consistent, likely because heuristics are indifferent to the goal of our analysis and uncertainty's role as signal or noise. Several heuristics are of particular importance for uncertainty visualisation. We will give a brief overview of them here, and for a more in depth analysis, we

recommend reading Padilla et al. (2022) as they provide a more in depth review.

The use of confidence intervals or boundaries have a few known issues in their interpretation. The sine illusion can cause the confidence interval of a smoothed sine curve to seem wider at the peaks than the troughs, causing us to underestimate uncertainty associated with changing values (Vanderplas & Hofmann 2015). Similarly researchers frequently find that visual boundaries such as the cone of uncertainty in hurricane forcasts (Padilla et al. 2022), or confidence bands or ellipses (VanderPlas & Hofmann n.d.) create the impression of a meaningful bound, separating points outside the bounds from those within it which can create problems if the bound is not inherently meaningful.

The choose of geometry is not neutral regarding the interpretation either, and several approaches have This can be considered an extension of the within bar bias, where participants looking at a bar chat with error bars view outcomes within the bar as more likely than those outside it (Newman & Scholl 2012).

While many researchers tend to prefer frequency framing, they are not without their critiques. Points that were on an outcome of an ensemble display were perceived as more likely than points not on an outcome, even when the point that was not on an outcome was closer to the centre of the distribution (and therefore more likely) (Padilla et al. 2017).

The final heuristic surrounds the finding that people seem to ignore uncertainty information if you give them the ability. Several studies have found that including the estimate on an uncertainty visualisation plot can cause viewers to ignore the uncertainty information and simply compare the difference between the two means (Padilla et al. 2022). These heuristics have the potential to create noise in signal suppression evaluations, and should be kept in mind when designing uncertainty visualisations as well as evaluation experiments.

5 Conclusions and Future work

This paper has identified gaps in the uncertainty visualisation literature that must be filled for the field to progress. We formalised the uncertainty visualisation problem, and in doing so highlighted an untouched area of uncertainty visualisation research.

Experimental practices on uncertainty visualisation need to be standardised. Uncertainty visualisations for decision making treat uncertainty as signal, while visualisations for signal-suppression treat uncertainty as noise. As the literature currently exists, there is no way to combine papers to get a meaningful sense of how uncertainty information is understood by a viewer. Researchers need to ensure that when they identify the motivation behind their visualisation technique, and that their evaluation methods match the motivations of the paper. Additionally *evaluation methods that evaluate uncertainty as noise need to be developed.*

Experimenters should consider evaluating visual aesthetics on how well they suppress information in a graphic. Research into separability and integrability is of particular interest to uncertainty visualisation, as it allows interference from the uncertainty variable. When designing experiments, authors often choose aesthetics that are visually distinguishable; uncertainty visualisation authors should consider doing the opposite.

Software that allows users to easily perform signal suppression should be developed. Existing uncertainty visualisation methods view a distribution as its own object and there are no software options for the "an uncertainty visualisation is a function of an existing visualisation" philosophy.

Signal suppression is an untouched area of visualisation research and developing methods for the practice may require us to challenge our entire notion of what makes a good visualisation.

We are interested to see how this challenge plays out.

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